

Architecture Justification

Food Bank Surplus Redistribution Platform

Architecture Selection: We chose Layered Architecture for the Food Bank Surplus Redistribution Platform.

Architectural Choice and Reasons

The Food Bank Surplus Redistribution Platform is organized using a Layered Architecture consisting of a Presentation Layer, Business Logic Layer, and Data Layer. The Presentation Layer contains the Web/Mobile UI Component used by the Donor Partner and Shelter Coordinator. The Business Logic Layer contains the Donation & Matching Service, Shelter Verification Service, and Notification Service, while the Data Layer contains the Database Service.

First, **Layered Architecture provides clear separation of concerns**. User interactions such as posting surplus food, reserving pickup windows, and viewing food listings are handled by the presentation layer, while matching food batches with shelters, verifying shelters, and sending notifications are handled within the business layer. This makes the system easier to understand and maintain.

Second, the architecture is suitable because the platform needs to **manage different types of functionality and data in an organized manner**. Food listings, dietary tags, expiry information, shelter details, and reservation information can be stored in the data layer, while business rules such as matching and verification remain in the business layer. This separation allows changes to one layer without unnecessarily affecting the others.

Security Advantage

The layered structure provides a degree of **separation between the user interface, business logic, and stored data**. Access to the database can be controlled through the business logic layer instead of allowing the user interface to access stored information directly. This helps protect sensitive platform data and keeps data access controlled through defined services.

Performance Benefit

The architecture can provide efficient processing by keeping related operations within the appropriate layer. The **Business Logic Layer** can process food matching and reservation operations without exposing database operations directly to users. In addition, the Notification Service can handle notification-related processing separately, supporting the requirement for instant notifications to shelters within a 5 km radius when high-quantity perishable food is listed.

Conclusion

Overall, Layered Architecture is an appropriate choice because it provides a clear and manageable structure for the Food Bank Surplus Redistribution Platform while supporting separation of responsibilities, controlled data access, and organized processing of food listings, shelter verification, matching, and notifications.